window of the vault. The locksmith would unlock it with the key kept inside the vault and return the box.

Since the vault had no door for a thief to enter and the second key was always inside the vault, it was never stolen. And this method worked very well to carry on trading securely.

Let's move to the present digital era. The important valuables being traded today are data. It is imperative that sensitive data be secured between the sender and the recipient. The lockbox, which is a standard one, represents a public key cryptosystem that is used to keep data secure. The two keys are public and private keys. Data encrypted with the public key can only be decrypted by the private key and vice versa.

There are multiple public key cryptosystems, the most notable ones being RSA (named after its creators Rivest, Shamir and Adleman), and ECC (elliptic curve cryptography). These provide mathematical functions that use a key to convert the data into some unreadable format which can be read only with the help of the other key. The keys are just one or a set of huge numbers which when passed to the mathematical function give the desired results.

We are not going into the details about the mathematics in this article but here is a representation of these systems:

- Encryption (Cleartext, Key1) → Ciphertext
- Decryption (Ciphertext, Key2) → Cleartext

Here, the cleartext is the original data while ciphertext is jargon that cannot be read by anyone without decrypting.

The public keys are usually announced for anyone to be able to use them. And the private keys are securely stored on the disk. This is how a sender can send valuable data to someone:

- Get the announced public key of the recipient Key1.
- Encrypt the cleartext with Key1 to get the ciphertext.
- Send the ciphertext.
- The recipient uses private key Key2 to decrypt the ciphertext and get the cleartext.
- Nobody other than the recipient has the private key so no one else can read it.

Note that it is important for the user to keep the private key securely. If anyone else gets that key, it may compromise the security of the data.

However, even if the private keys are stored securely on the disk, malicious applications and viruses are today able to retrieve them on the disk or memory, compromising the data. It is now understood that storing the keys on a disk is not a great idea since these can always be accessed by malware.

That's how the concept of building a vault in the basement has come up. The basement is not a part of the general living space in a home. And the small window does not allow any thief to go in. Similarly, in the modern digital era, a new chip called a trusted platform module (TPM) has been developed. It is not a part of the disks, the CPU or the memory, all of which can be accessed by malware. It is an entirely different chip placed on the motherboard.

Just like the locksmith stayed in the basement and made keys there, the TPM allows us to create keys inside it and take out only the public key. The TPM never allows the private keys to leave it. If any operation is to be performed with the private key, the application sends the ciphertext to the TPM, which decrypts it and returns the cleartext.

Figure 2: A trusted platform module (TPM)

This ensures that the private keys are never compromised.

But what about devices that do not have a TPM? Well, some mobile devices do not have it, but they can use something called the TEE (trusted execution environment). It partitions a part of the CPU and the disk to behave like a TPM, and the operating system ensures that no application has access to the keys stored on the TEE. But it is not as secure as having a physical TPM.

Some operating systems like Windows 11 now compulsorily need the computer to have a TPM to be able to function securely and correctly. This ensures the user's data is never compromised. Apple devices come with an in-built similar chip called Secure Enclave, Samsung devices come with Knox, and Google devices come with Titan. These ensure that the private keys are protected against highly advanced cyberattacks even if the device is lost. END

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